$$Ad_2 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} = 0\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + 3\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + 0\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} \quad \text{(By defn. of } Ax\text{)}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 3 \\ 6 \\ 12 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{(By defn. of scalar multiple)}$$

$$= \begin{bmatrix} 3 \\ 6 \\ 12 \end{bmatrix} \quad \text{(By defn. of vector sum)}$$

The 2nd column of AD is a linear combination of the columns of A where the weights are the corresponding entries of the second column of D.

$$Ad_3 = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} = 0\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + 0\begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} + 5\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix} \quad \text{(By defn. of } Ax\text{)}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 5 \\ 15 \\ 25 \end{bmatrix} \quad \text{(By defn. of scalar multiple)} = \begin{bmatrix} 5 \\ 15 \\ 25 \end{bmatrix} \quad \text{(By defn. of vector sum)}$$

The 3rd column of AD is a linear combination of the columns of A where the weights are the corresponding entries of the third column of D.

$$AD = \begin{bmatrix} 2 & 3 & 5 \\ 2 & 6 & 15 \\ 2 & 12 & 25 \end{bmatrix} \quad \text{(By defn. of matrix multiplication)}$$

$$DA = \begin{bmatrix} Da_1 & Da_2 & Da_3 \end{bmatrix}$$

$$Da_1 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 1\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + 1\begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} + 1\begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} \quad \text{(By defn. of } Ax\text{)}$$

$$= \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} \quad \text{(By defn. of scalar multiple)}$$

$$= \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix} \quad \text{(By defn. of vector sum)}$$

The 1st column of DA is a linear combination of the columns of D where the weights are the corresponding entries of the 1st Column of A.